

β Canis Majoris photosphere pulsation modes: Observation with Veritas Stellar Intensity Interferometry

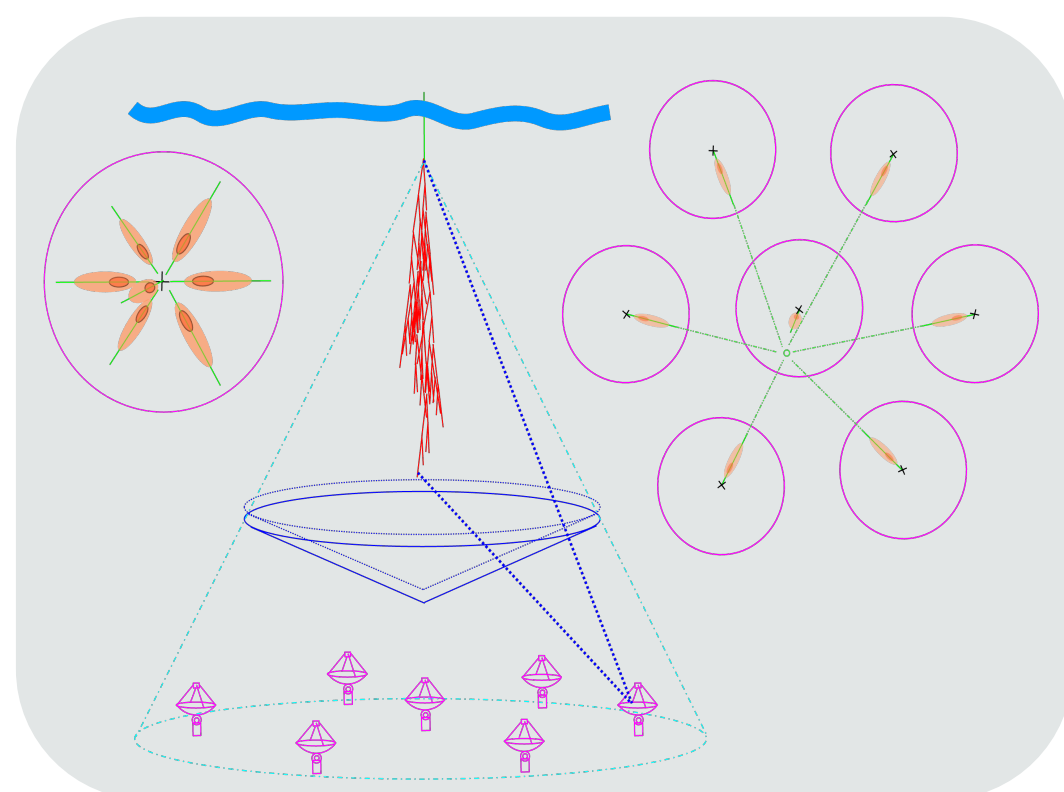
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β Canis Majoris

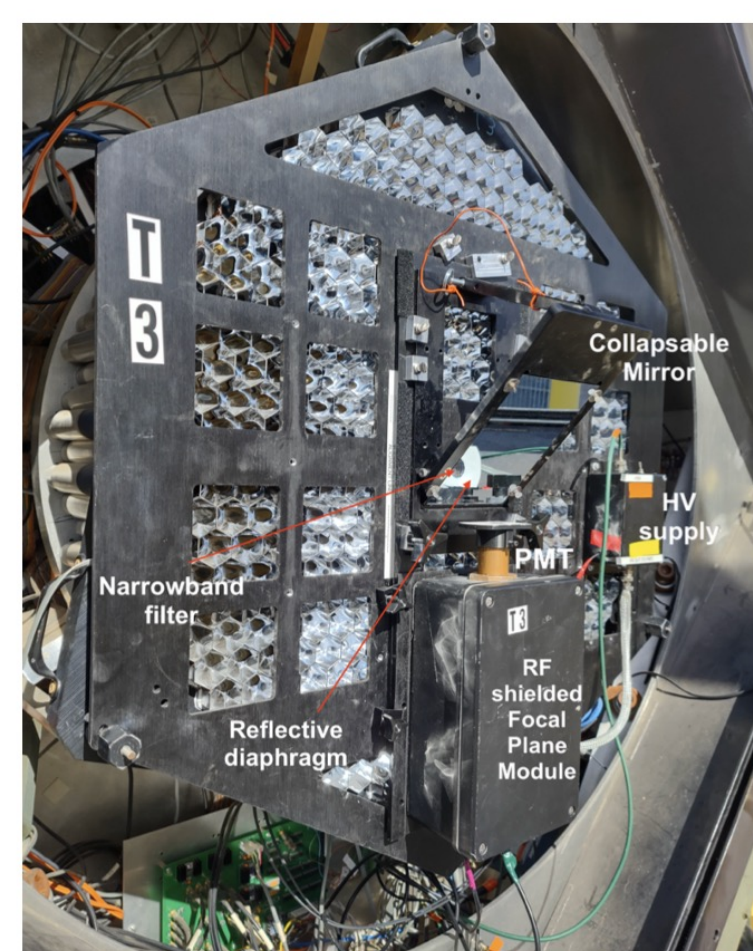
- B1 II-III type star
- Apparent Magnitude varies from +1.97 to +2.01
- Distance from Hipparcos: (151 ± 0.5) pc
- Pulsation periods: around 6 hours
- Rotation Period: 13.6 days^[1]
- Inclination as seen from Earth: 57.6° ^[2]
- Radii given by two models: $(7.4 \pm 0.85) R_\odot$ ^[3], $(8.2 \pm 0.55) R_\odot$ ^[4]
- Expected range of Angular Diameter: 0.456mas - 0.505mas

The VERITAS Array

VERITAS (Very Energetic Radiation Imaging Telescope Array System) is designed as an Imaging Atmospheric Cherenkov Telescope array used for gamma-ray astronomy by detecting gamma ray induced extensive air showers.

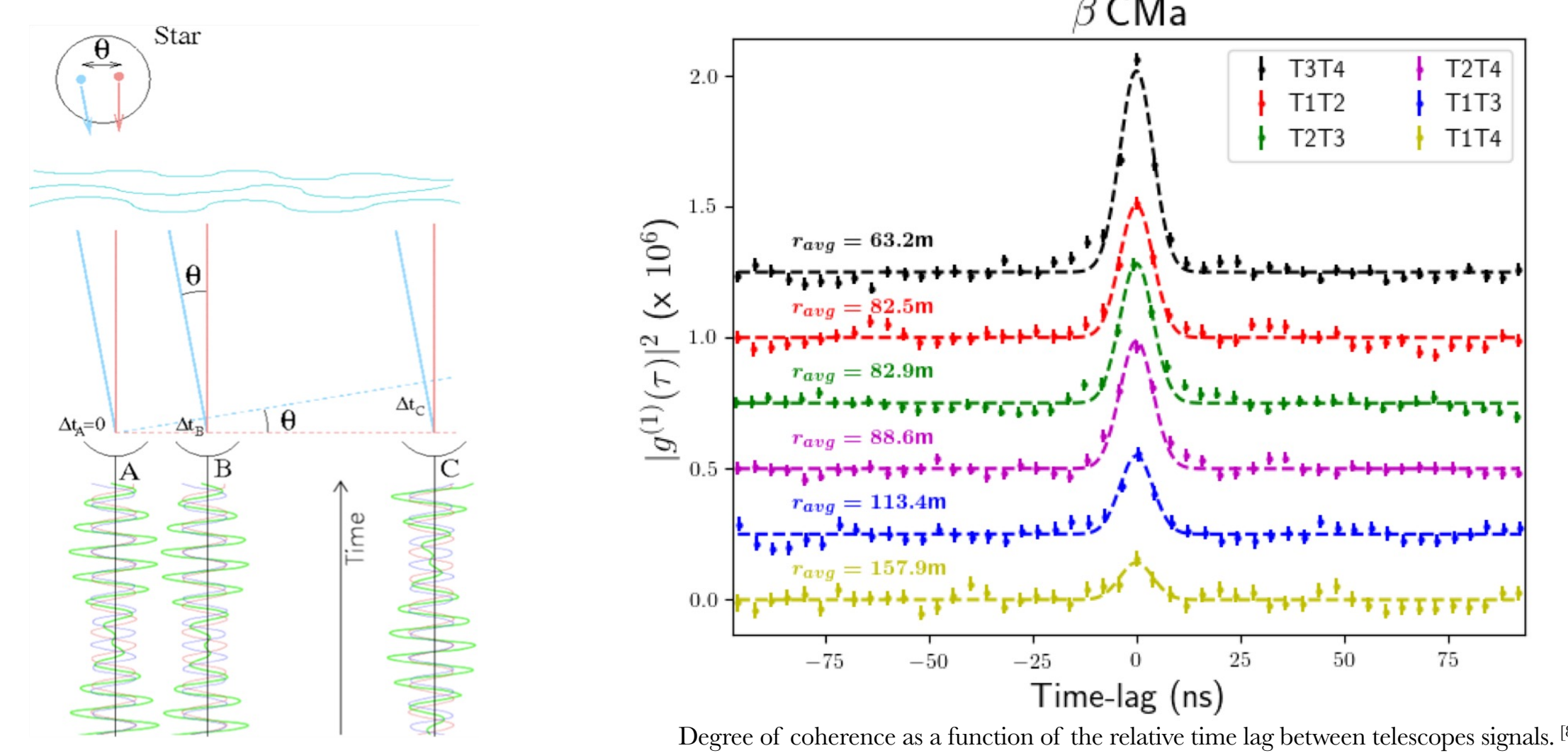


The VERITAS telescopes are also used as light receivers for Stellar Intensity Interferometry (VSII mode) at wavelength (416 ± 5) nm^[5], during bright-moon periods when gamma-ray observations are not feasible.^[6]



Stellar Intensity Interferometry

Developed by Robert Hanbury Brown in 1963, Stellar Intensity Interferometry relies on the fact that light from a star received by two separate photoelectric detectors produces correlated current fluctuations Δi_1 and Δi_2 .^[7]



The correlation is proportional to the squared degree of coherence, or squared visibility $|\gamma_d^2|$ ^[9]:

$$\frac{\langle \Delta i_1(t) \Delta i_2(t) \rangle}{\langle i_1 \rangle \langle i_2 \rangle} = |\gamma_d^2|$$

The degree of coherence depends on the relative positions of the detectors and on the angular distribution of the light source. For a uniform disk model with angular diameter θ_{UD} and baseline b ^[10]:

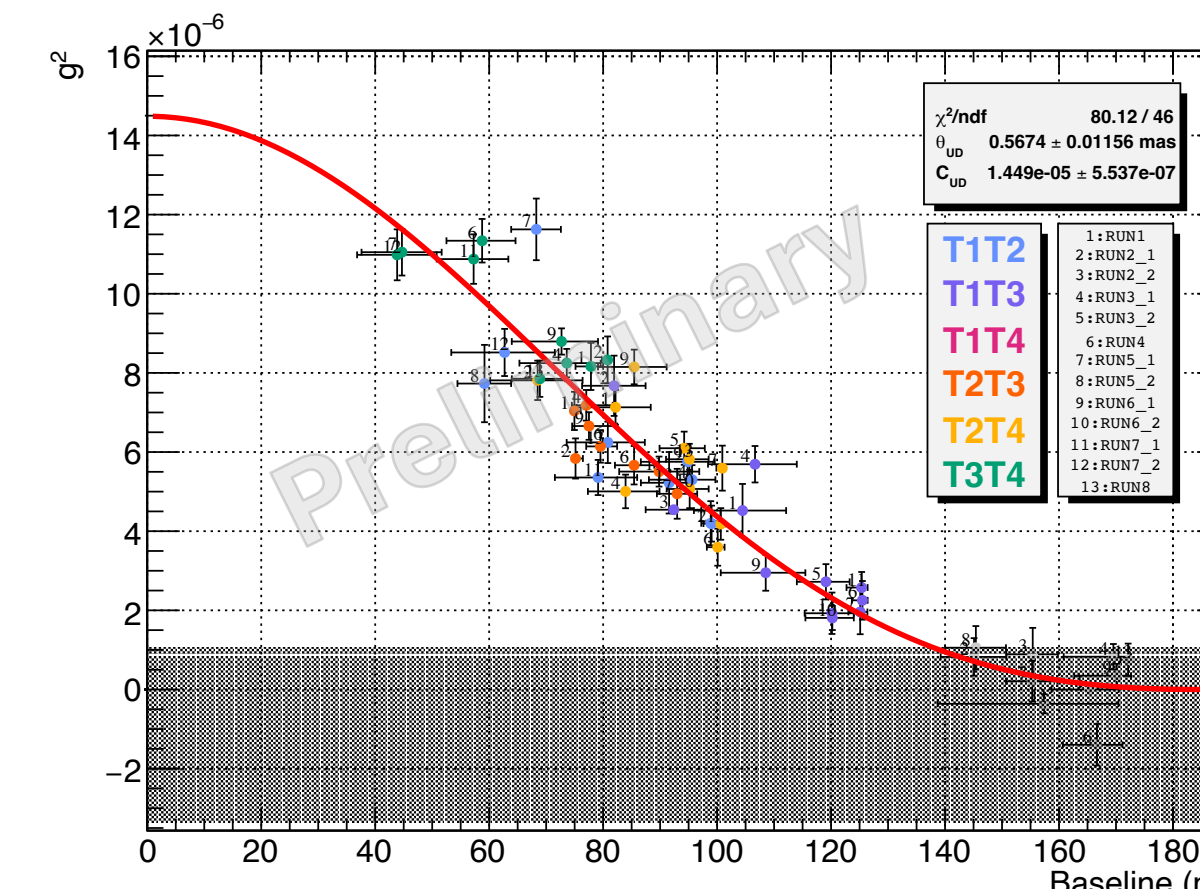
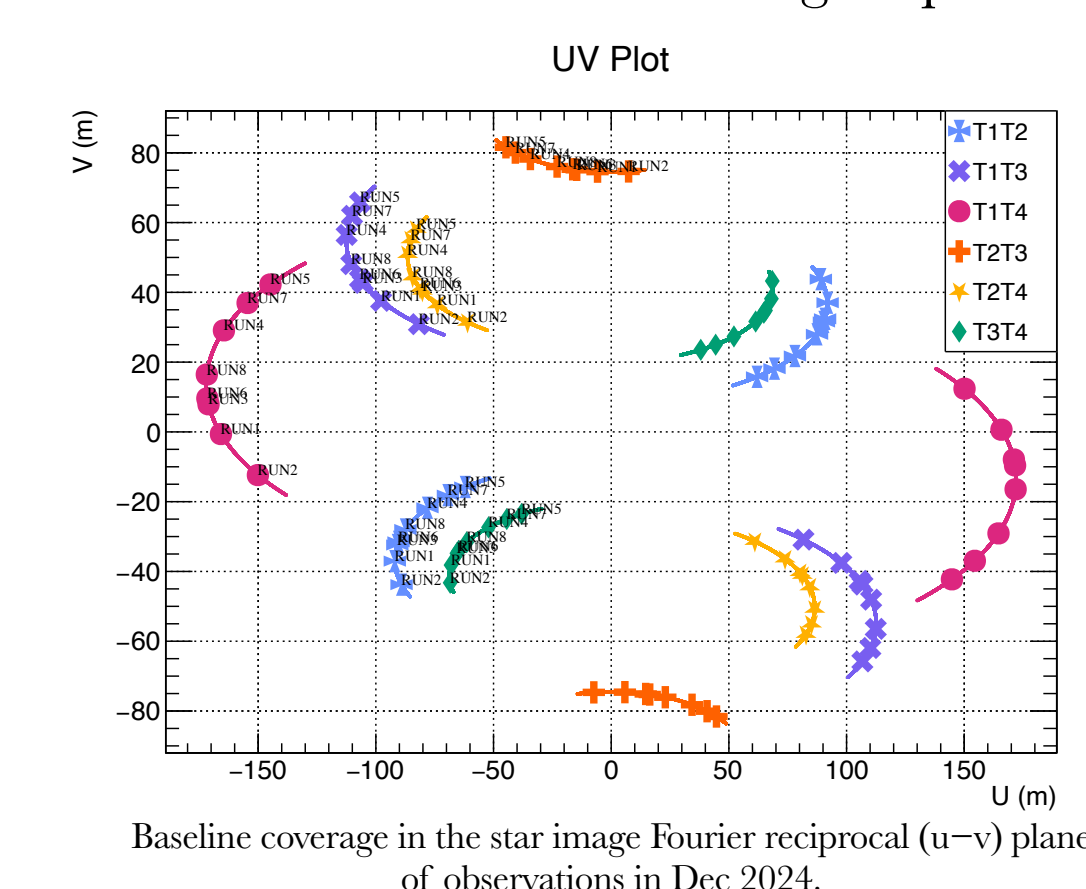
$$\gamma_d^2(\theta_{UD}, b) = C_{UD} \times \left(\frac{2J_1(\pi b \theta_{UD} / \lambda)}{\pi b \theta_{UD} / \lambda} \right)^2$$

This can be used to measure the size and shape of bright stars.

Mean Angular Diameter

β Canis Majoris was observed with VSII for 70+ hours from 2019 to 2025. Here we report our preliminary analysis of the Dec 2024 data.

By fitting a uniform disk model to the data, we obtain a preliminary angular diameter measurement of (0.57 ± 0.012) mas confirming MAGIC's result^[11] (0.545 ± 0.046) mas with both measurements exceeding expectation from Brott's model^[4].

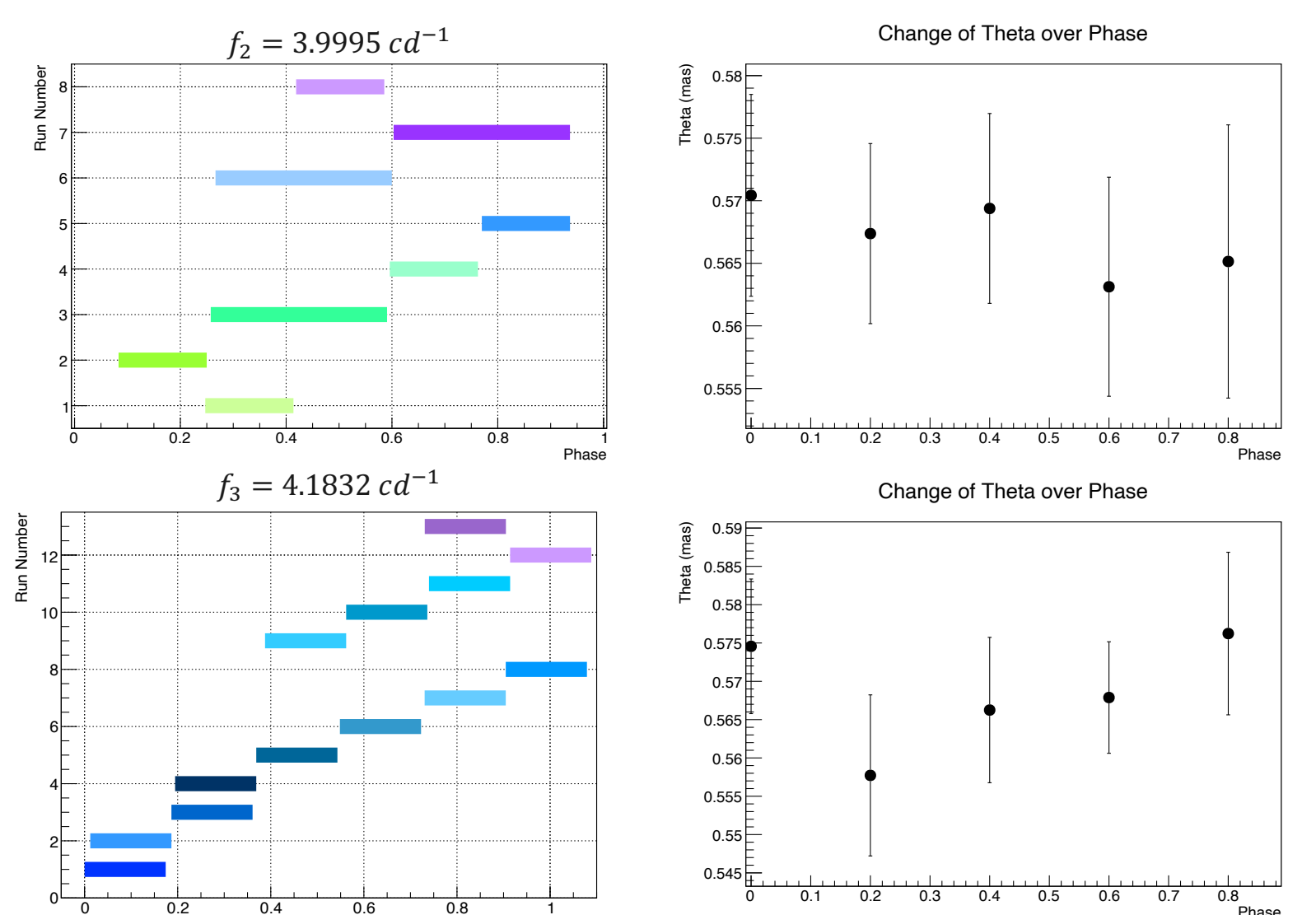
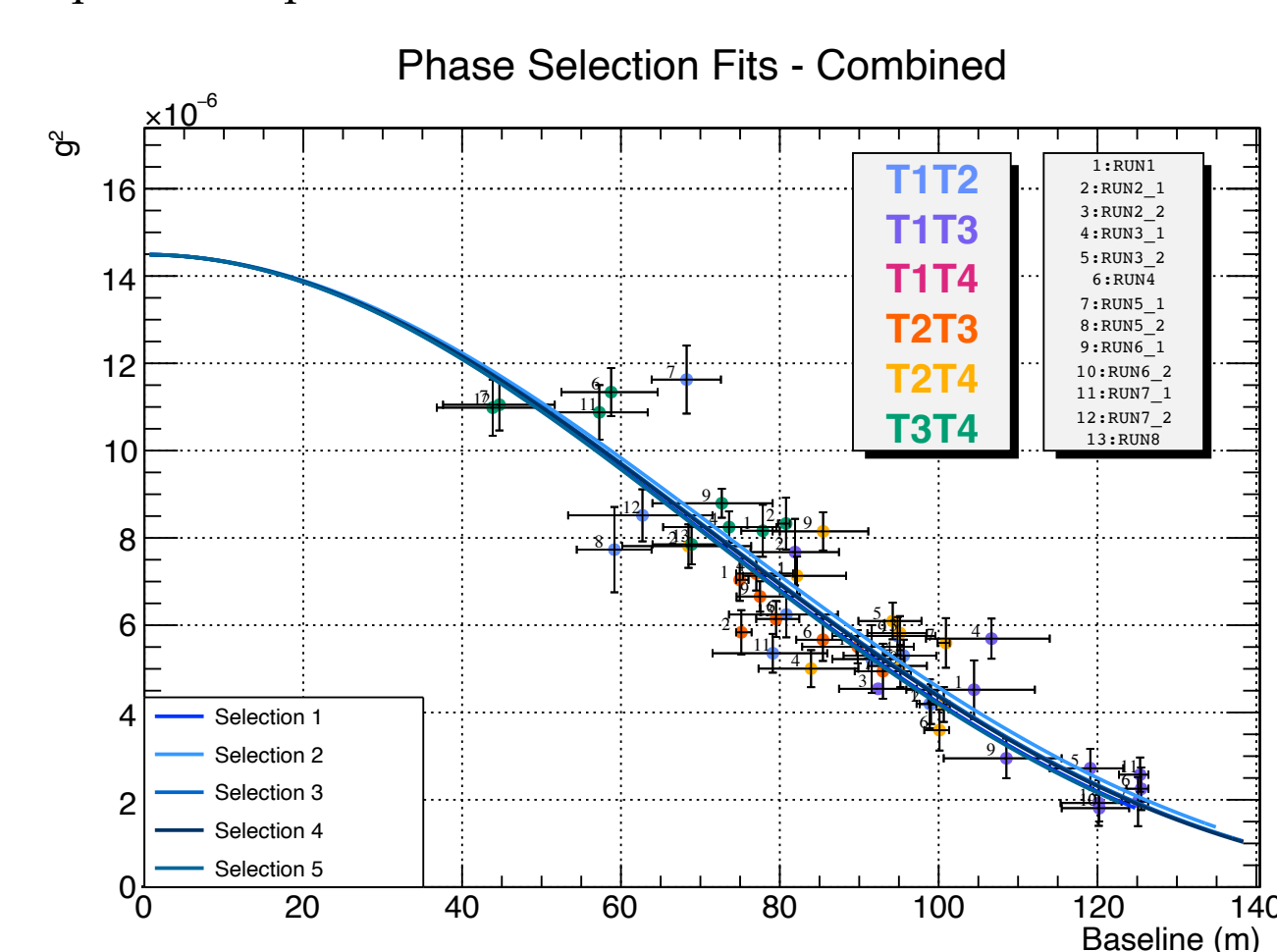


Search for Modulation

There are three well-established pulsation frequencies^{[12] [13]}:

$$\begin{aligned} f_1 &= 3.9793 \text{ day}^{-1} \text{ Non-radial } (\ell = 2) \\ f_2 &= 3.9995 \text{ day}^{-1} \text{ Radial } (\ell = 0) \\ f_3 &= 4.1832 \text{ day}^{-1} \text{ Non-radial } (\ell = ?) \end{aligned}$$

We fold our observations by the corresponding periods, allowing features to align coherently at each pulsation phase.



We fit a uniform disk model to each phase selection with C_{UD} as a fixed parameter obtained from the mean angular diameter measurement. The resulting angular diameters are then plotted as a function of phase. Given the uncertainties with the data presented here, we could only report upper limits on the pulsation amplitudes.

Conclusion

Stellar photosphere pulsations can be detected with VSII.

Ongoing work focuses on:

- Reducing statistical uncertainties by incorporating additional data from previous observations.
- Accounting for systematic effects.

In parallel, we are investigating modeling approaches, including:

- Multi-component disk model
- Non-radial pulsation models

These efforts aim to provide insight into the star's internal oscillation modes, complementing results from asteroseismology.

Acknowledgments

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